

Varshil Shah

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SUMMARY

AI Engineer delivering production LLM applications and Retrieval-Augmented Generation (RAG) systems. Experienced in Python, agent orchestration, API integration, and AWS, with a focus on monitoring, debugging, reliability, and human-in-the-loop controls.

TECHNICAL SKILLS

Languages & APIs: Python, JavaScript, SQL, REST APIs, FastAPI, Flask

AI & ML: LLMs, Agentic AI, RAG, Agent Orchestration, Prompt Engineering, Model Evaluation

Frameworks & Data: LangChain, MCP, FAISS, MongoDB, Snowflake, Databricks, Elasticsearch/Kibana

Cloud & Delivery: AWS, Docker, Kubernetes, Event-Driven Architecture, Airflow, CI/CD, Git

PROFESSIONAL EXPERIENCE

TD SYNnex Corporation

Gilbert, AZ

AI Business Systems Engineer II

May 2025 – Present

- Designed and deployed enterprise LLM email triage; co-designed reusable orchestration and document-processing components integrating Microsoft Graph, structured extraction, and ticketing APIs. Scoped **3,000 plus mailboxes** with **\$43.75M modeled annual labor value** at full adoption.
- Cut measured inference costs **86–88%** (approximately \$1,498 per project per month) through context engineering, deterministic prefilters, and task-specific agents; reduced email inputs from **29K to 620 tokens/call**.
- Translated business requirements into multi-step JavaScript agent workflows with API tool calls, structured outputs, and data contracts; implemented reply-loop prevention, duplicate-ticket controls, and human-review safeguards.
- Built shared Python/FastAPI monitoring with MongoDB checkpoints and Elasticsearch/Kibana telemetry for decision tracing, debugging, and reliability analysis; evaluated classifications against keyword evidence in shadow mode.
- Standardized AI-assisted development with **7 Cursor skills**, Model Context Protocol (MCP) integrations, and reusable code-generation templates; led weekly cross-functional training for new use cases.

Data Engineering & Analytics Intern

May 2024 – Jan 2025

- Built knowledge-grounded RAG for text-to-SQL using semantic models, knowledge graphs, and Snowflake metadata, improving query accuracy by **40%**.
- Improved retrieval efficiency **30%** using LLM fine-tuning, active learning, semantic caching, FAISS vector indexing, and query batching; integrated OpenAI GPT, Databricks, and Snowflake Cortex.

Arizona State University

Tempe, AZ

Graduate Research Associate

Oct 2023 – May 2025

- Processed **10 plus TB of imagery** with Planet APIs and Meta SAM for urban-development analysis, improving processing efficiency by **15%**.
- Benchmarked GPT-4 predictions from Street View imagery and census data against segmentation models; evaluated demographic bias through controlled image interventions and regression-based Responsible AI analysis.

Anicca Data Science Solutions Ltd

Bangalore, India

Data Engineer

Aug 2022 – Jul 2023

- Developed AWS Glue/PySpark ETL for **millions of ERP records**, using Kinesis, Firehose, and Lambda for event-driven ingestion, fault tolerance, and data-quality checks.
- Saved **\$20K/month** across EC2, S3, Airflow, and Redshift data workloads; shortened computation time by **70%** through transformation, modeling, and partitioned storage.
- Migrated workloads to AWS and built CI/CD pipelines with unit and integration tests for production releases.

EDUCATION

Arizona State University — Master of Computer Science; GPA: 3.97/4.00

May 2025

Ahmedabad University — B.Tech., Information and Communication Technology

May 2022

PROJECTS

Reinforcement Learning for Algorithmic Paper Trading — Python, FinRL, PPO

Applied Proximal Policy Optimization (PPO) to DOW 30 paper trading via Alpaca; grew a simulated \$1M to \$1.084M.

PUBLICATION

Performance Comparison of Deep Neural Networks for classification of Indian Coins — IEEE Xplore

Achieved **97% classification accuracy** using transfer learning; evaluated 10 plus CNN architectures for coin recognition.